MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU- UGP (HONOURS)

FIFTH SEMESTER EXAMINATION

2024 ADMISSION ONWARDS

GENETICS MG5DSCBTS302

Total:70 marks

Time: 2 hours

Students should attempt at least one question from each course outcome to enhance their overall outcome availability.

Part A

Multiple Choice Questions

Answer all questions, each question carries 1 mark

1.Which cross is used to identify the genotype of a plant. [K] [CO2]

[A] Test cross [B] Reciprocal cross [C]Back cross [D] None of the above

2. What is the phenotypic ratio of a typical Mendelian cross. [U] [CO2]

[A] 9:3:3:1 [B] 3:1 [C] 1:2:1 [D] Both A and B

3. Pick the odd one out [K] [CO1]

[A] Flower colour [B] Seed colour [C] Seed shape [D] Shape of the pollen

4.What is the percentage of off springs with the genotype tt, in the cross Tt x Tt [U] [CO1]

[A] 25 [B] 50 [C] 75 [D] 100

5.Plumage colour in white leghorn is due to [K] [CO4]

[A] Recessive epistasis[B] Lethality [C] Pleiotropy [D] Dominant epistasis

6.Who discovered the phenomenon linkage? [K] [CO4]

[A] T.H.Morgan [B] C.B.Bridges [C] A.H. Stuartevant [D] Bateson and Punnet

7.Male honey bees are [K] [CO3]

[A] Haploid[B]Diploid [C] Triploid [D] None of the above

8. How many linkage groups are there in humans[U] [CO1]

[A] 23 [B] 46 [C] 22 [D] 44

9.In human karyotype, the number of autosomes [U] [CO4]

[A] 44 [B] 22 [C] 23 [D] 46

10. ZW-ZZ system is seen in [K] [CO4]

[A]Chickens [B] Butterflies [C] Drosophila [D] Grass hopper

Part B

Short answer questions

Answer any 10 questions, each question carries 3 marks

11. Differentiate genotype and phenotype [U] [CO2]

12.What are reciprocal cross and back cross. [U] [CO1]

13.What are complementary genes? Give an example [K] [CO2]

14.Write briefly on lethal genes with an example. [U] [CO1]

15.What is genic balance theory? [U] [CO3]

16.Explain environmental influence on sex determination. [U] [CO1]

17. Differentiate sex linked inheritance, sex limited characters and sex influenced characters. [U] [CO1]

18.Explain polygenic inheritance with an example. [U] [CO1]

19.What are the characteristics of multiple alleles? [U] [CO1]

20.Define Barr body. What are free martins? [K] [CO2]

21.What is Lyon hypothesis? [U] [CO2]

22 State Mendel's laws of independent assortment. [K] [CO4]

Part C

Short essay questions

Answer any 6 questions, each question carries 5 marks

23.Describe test cross with an example. [U] [CO1]

24.Explain dominant and recessive epistasis with examples. [K] [CO2]

25.Explain multiple allelism with an example. [U] [CO1]

26. Describe in detail the phenomenon linkage with an example. [K] [CO3]

27.Describe in detail human karyotyping. [U] [CO3]

28. Explain in detail inborn errors of metabolism. [K] [CO4]

29.Explain the different theories of sex determination. [U] [CO4]

30. Explain in detail the different chromosomal abnormalities. [K] [CO4]